AI: Introduction to Keras and Our First Data Set

In the last article in this series on AI and machine learning, we learned to use Anaconda. We also tried to strengthen our knowledge of probability, a task we will continue in this article. We will discuss seaborn and Keras, and further explore Pandas and TensorFlow in this sixth article in the AI series.

Let us begin by increasing our theoretical knowledge of AI and machine learning. We all know that AI, machine learning, data science, deep learning, etc, are the trending topics in computer science today. However, there are other topics trending in this domain too such as blockchain, IoT (Internet of Things), quantum computing, etc. So do the advances in the field of AI affect any of these technologies in a positive way?

First, let us discuss blockchain. According to Wikipedia, "A blockchain is a type of distributed ledger technology that consists of a growing list of records, called blocks, that are securely linked together using cryptography." At a casual glance it looks as if AI and blockchain are two independent technologies advancing at a very high pace at the same time in history. But surprisingly, this is not the case. The buzzword associated with blockchain is integrity and the buzzword associated with AI is data. We give massive amounts of data to AI based applications to process. Though the results produced by many of these applications are stunning, how do we trust them? This raises the need for explainable AI, which can give certain guarantees so that end users can trust the results provided by AI based applications. Many experts believe that blockchain technologies can be used to increase the trustworthiness of decisions made by AI based software. On the other hand, smart contracts (which are part of blockchain technology) can benefit from AI based verification. Notice that, in essence, smart contracts and AI often lead to decision making. Thus, we can safely conclude that the advances in AI will positively affect blockchain technology and vice versa.

Now, let us discuss how AI and IoT impact each other. In the earlier days, IoT devices often did not have large processing power or battery backup. This made the deployment of machine learning based software, which requires large processing power, infeasible in many IoT devices. At that point in time, only rule-based AI software was being deployed in most IoT devices. The advantage with rule-based AI is that it is simple and requires relatively less processing power. However, nowadays, IoT devices are equipped with more processing power and can run powerful machine learning software. The advanced driver-assistance system developed by Tesla, called Tesla Autopilot, is an excellent example for the blending of IoT and AI. Thus, here again, AI and IoT are positively affecting each other's development.

Finally, let us try to understand how AI and quantum computing are impacting each other. Even though quantum computing is still in its infancy, quantum machine learning (QML) is taken very seriously. Quantum machine learning is based on two concepts: quantum data and hybrid quantum-classical models. Quantum data is the data generated by